VIGIL: A Validation-First Research Infrastructure for Global Infectious-Disease Intelligence

Yen-Hsiang Wang, MD, MSc

Graduate Institute of Medical Sciences, College of Medicine, Taipei Medical University, Taipei 11031, Taiwan

rogerwang890928@gmail.com · ORCID 0000-0003-0307-8447

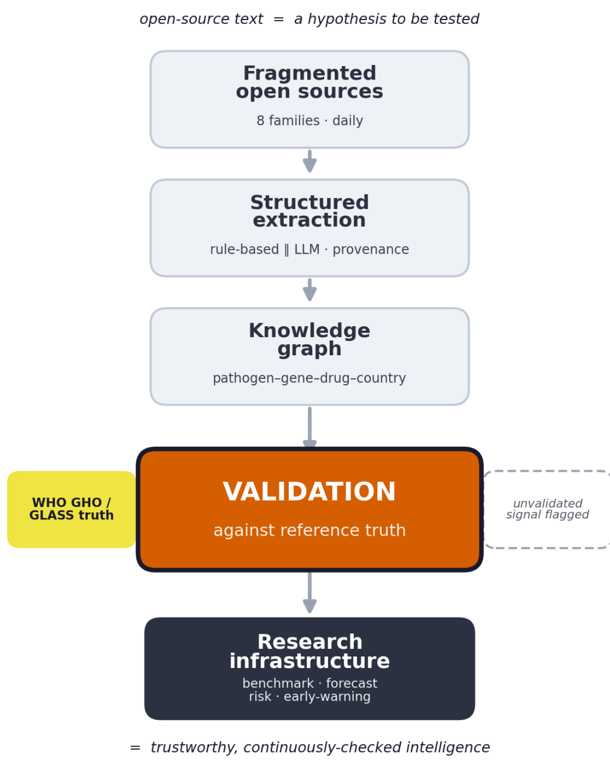

Figure 1. VIGIL as a validation-first research infrastructure. Open-source text enters as fragmented signal; it is extracted to a common schema and linked into a knowledge graph, and then — the pivotal step — *validated against independent WHO GHO/GLASS reference truth* before it is trusted. Validated signal becomes research infrastructure (benchmarking, forecasting, risk, early-warning); unvalidated signal is flagged, not acted on.

Abstract. Infectious-disease and antimicrobial-resistance (AMR) intelligence is increasingly mined from open-source text — outbreak alerts, the peer-reviewed and preprint literature, and trial registries. But text-derived signals inherit reporting and publication bias, and are almost never checked against independent reference truth before they inform situational awareness. We argue that open-source infectious-disease intelligence should be treated not as a signal to be trusted but as a hypothesis to be *validated*, and that this validation must be continuous, reproducible, and grounded in reference surveillance data. *How should open-source infectious-disease intelligence be validated?* We answer with **VIGIL**, a validation-first research infrastructure that collects documents daily from eight open source families, extracts a fixed schema with a transparent rule-based baseline and a large language model (LLM) side by side, links entities into a knowledge graph, and — as its primary operation — grounds every derived signal against WHO GHO/GLASS reference resistance data rather than asserting it. From a snapshot of 2,075 documents VIGIL produced 31 pathogens, 12 resistance mechanisms, 62 countries, and 1,060 knowledge-graph relations, aligned against 1,005 reference resistance values. Validation immediately proves its worth: across 43 countries, raw literature volume was essentially uncorrelated with measured MRSA bloodstream-infection resistance (Spearman $\rho = 0.07$), and even after publication-normalisation the association remained weak ($\rho = 0.16$) — direct evidence that text volume must not be read as a resistance proxy. Rather than a weakness, this result is the point: VIGIL turns an unexamined caveat into a measurable, continuously monitored target, and provides the standing apparatus for a program of validation studies (LLM-extraction benchmarking, lead-time forecasting against GLASS, and risk calibration). It is open, runs and updates daily for free, and is archived with a citable DOI.

Keywords: validation-first surveillance; external validation; research infrastructure; antimicrobial resistance; large language models; information extraction; knowledge graph; open data; Asia-Pacific.

Availability: Code github.com/Rogerking928/id-intel-platform · Archive/DOI 10.5281/zenodo.21263882 · Live demo streamlit.app · Licence MIT.

1 Introduction

Knowing where resistant pathogens and outbreaks are emerging is a fundamental problem for global health, and antimicrobial resistance (AMR) — associated with millions of deaths annually and heaviest where surveillance is weakest [1] — makes it acute. The evidence is not missing; it is scattered. Every day, national and supranational alerts, the peer-reviewed and preprint literature, and clinical-trial registries emit open-source text about resistant organisms, in heterogeneous formats and without a common structure. Converting this stream into intelligence a clinician or analyst can act on requires connecting the same entities — pathogen, resistance mechanism, antibiotic, disease, and place — across all of these sources, uniformly and continuously.

Extracting this structured tuple at scale requires three things together: broad automated ingestion of open sources, uniform entity–relation extraction, and grounding against reference surveillance data. Prior work supplies each ingredient in isolation, but none supplies all three. Open-source epidemic-intelligence systems — WHO EIOS, HealthMap, ProMED-mail, EPIWATCH — aggregate and flag disease events well [3, 4, 5, 6], but they target outbreak *events* rather than the *structured, AMR-specific* pathogen–gene–antibiotic–country relationships that AMR research and stewardship need, and several are proprietary or not straightforwardly reproducible. Curated surveillance systems such as WHO GLASS provide authoritative laboratory resistance data, but with reporting lag and limited country coverage [2]. Large language models (LLMs) now make structured extraction from free text feasible, but their use in AMR surveillance raises unresolved questions of accuracy, reproducibility, cost, and clinician trust, and there is little open, benchmarked tooling — particularly outside high-income, English-clinical-note settings.

Almost every system above produces a signal and asks the reader to trust it; few treat *validation against independent reference data* as a first-class, standing operation. We take this to be the decisive gap, and reframe the problem accordingly: *How should open-source infectious-disease intelligence be validated — continuously, reproducibly, and against reference truth?* Extraction accuracy is a means; validation is the object. We present **VIGIL** (Vigilance Intelligence for Global Infectious-disease & resistance Landscape), a validation-first research infrastructure built around this question: it ingests and structures open sources automatically, but every derived signal is grounded against WHO GHO/GLASS reference resistance before it is reported, and each analytic module is posed as a research question with an explicit validation design. VIGIL is released fully open-source, runs at no cost, and carries an Asia-Pacific lens.

This paper makes four contributions:

- **Validation as the primary operation.** VIGIL grounds every text-derived signal against independent WHO GHO/GLASS reference resistance data rather than asserting it, making validation — not extraction — the object

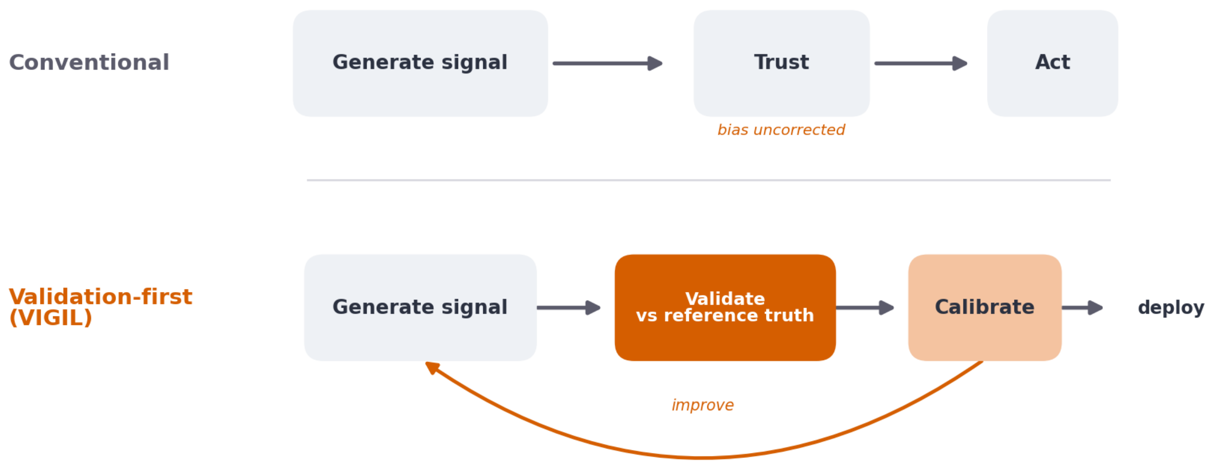

Figure 2. The design shift VIGIL embodies. Conventional open-source intelligence *generates a signal and trusts it*, carrying uncorrected reporting and publication bias into action. A validation-first system instead treats each signal as provisional: it is validated against independent reference truth, used to calibrate the method, and only then deployed — a continuous loop rather than a one-way pipeline.

of study.

- **An open, provenance-complete pipeline** that ingests eight open source families daily and records full provenance (raw text, source, timestamp, and, per extraction, the extractor, model, and prompt version), turning extraction itself into a reproducibly benchmarkable object with a transparent rule-based baseline and an LLM run side by side on identical inputs.
- **A knowledge graph and a suite of validation-first modules** (trends, forecasting, novelty, risk, outbreak, and an extraction benchmark), each framed as research question → validation design → current result rather than as a product feature.
- **Evidence that validation is necessary**, obtained by turning the infrastructure on itself: text volume is essentially uninformative about measured resistance — reframing an unexamined caveat into a measurable, continuously monitored target and a research program.

2 Implementation

2.1 Overview and architecture

VIGIL follows a linear, auditable pipeline: **collect** → **extract** → **entities/relations** → **analyse** → **serve** (Figure 1). Documents are stored in a single SQLite database. Every document retains its raw text, source, URL, and fetch timestamp; every extraction retains the extractor name, model version, prompt version, and raw output. This provenance makes each analytic result traceable to source text and supports head-to-head comparison of extractors.

2.2 Data sources

Eight source families are configured (Table 1), each accessed through public APIs or feeds and de-duplicated so that only new records are appended on each run. Sources span official alerts (WHO DON, CDC, ECDC, UKHSA), the literature (PubMed [8], bioRxiv/medRxiv via Europe PMC [9]), trials (ClinicalTrials.gov [10]), and reference resistance data (WHO GHO/GLASS [7, 2]).

Table 1. Configured data sources.

Source	Type	Access
WHO Disease Outbreak News	Official outbreak alerts	JSON (OData) API
US CDC	Alerts / MMWR / newsroom	RSS
ECDC	European communicable-disease news	RSS
UK Health Security Agency	News & guidance	GOV.UK Atom feed
PubMed	Peer-reviewed literature	NCBI E-utilities
bioRxiv / medRxiv	Preprints	Europe PMC REST API
ClinicalTrials.gov	Trial registrations	API v2
WHO GHO / GLASS	Reference resistance indicators	GHO OData API

2.3 Information extraction

Two extractors return an identical schema. The **rule-based** extractor applies frozen controlled vocabularies (a versioned codebook) with word-boundary matching, country-alias canonicalisation, and assertion/negation handling; it is free, offline, deterministic, and serves as the benchmark baseline. The optional **LLM** extractor (Google Gemini in the current build; the interface is model-agnostic) is prompted to return the same JSON schema. Because both outputs are stored per document, VIGIL supports a direct, reproducible comparison of methods — for example, on a paper describing altered ceftazidime-avibactam resistance, the LLM recovered specific alleles (KPC-234, NDM-1) and plasmid context that the dictionary baseline did not.

2.4 A suite of validation-first modules

Entity co-occurrence within a document generates Pathogen → Country → Gene → Antibiotic → Disease relations, queryable as a graph. On top of the graph and entity tables, VIGIL exposes its analytics not as product features but as a suite of modules, each posed as a *research question* with an explicit *validation design* and an honestly reported *current status* (Table 2). This framing is deliberate: it makes the infrastructure's job the continual production *and checking* of surveillance signals, and holds every module accountable to reference truth rather than to face validity.

Table 2. VIGIL's validation-first modules, each framed as a research question with a validation design and its current status. Status is reported honestly: several modules are infrastructure-ready but await accumulated history or expert labels.

Module	Research question	Validation design	Current status
Extraction benchmark	Do a transparent rule-based extractor and an LLM agree, and which is more accurate?	Clinician-annotated gold standard; per-field F1 and inter-annotator κ; frozen codebook and leakage controls	Infrastructure and `annotations` table ready; gold set not yet annotated
Construct validity (risk)	Does a text-derived AMR signal reflect *measured* resistance?	Cross-sectional Spearman vs WHO GHO MRSA / *E. coli* BSI resistance; raw vs publication-normalised	ρ = 0.07 raw, 0.16 normalised across 43 countries (Fig. 6)
Outbreak	Does literature signal anticipate *officially reported* outbreaks?	Source-separated: outcomes from official alerts, features from literature; rank-AUC	Qualitative negative; precise AUC to be re-run on the current snapshot
Forecasting	Is discussion volume predictable, and does it lead reference surveillance?	Transparent baseline + rolling backtest; planned lead-time evaluation vs GLASS	Prototype with backtest; lead-time pending longer history
Novelty	Which entity combinations are genuinely new?	Flag combinations never previously co-occurring in the dated corpus	Implemented; precision pending expert review
Trends / emerging signal	What is rising fastest, and where?	Growth-rate and emerging-signal detection over the dated corpus	Implemented; preliminary (young corpus)

2.5 Validation design

Analytics are grounded in reference data rather than asserted. WHO GHO/GLASS resistance indicators (proportion of MRSA and third-generation-cephalosporin- resistant *E. coli* bloodstream infections) are ingested automatically as ground truth. Construct validity is assessed cross-sectionally against these values. The outbreak task deliberately separates sources — outcomes are taken from official alerts (WHO/CDC/ECDC/UKHSA) while features derive from the literature stream — to avoid circularity. A frozen annotation codebook and an `annotations` table support a planned clinician-annotated extraction benchmark under leakage controls.

2.6 Availability and reproducibility

VIGIL is open-source (MIT), archived on Zenodo with a DOI, deployed as a public live instance (Appendix A), and updated daily at no cost via continuous integration. Code, the codebook, and the schema are versioned; a single command reproduces the pipeline.

3 Results

3.1 Corpus

The analysed snapshot contained 2,075 extracted documents. Seven sources contributed actively (CDC n = 1,828; preprints n = 116; PubMed n = 54; ClinicalTrials n = 37; UKHSA n = 20; ECDC n = 10; WHO DON n = 10; Figure 3a), spanning research articles, outbreak alerts, AMR reports, guidelines, and trials (Figure 3b); the GLASS ingestion path is included but was not populated in this snapshot. Extraction yielded 31 distinct pathogens (including the priority phenotypes MRSA, VRE, CRE, CRAB, and *Candida auris*), 12 resistance genes/mechanisms, 23 antibiotics, 62 countries, and 1,060 knowledge-graph relations. The WHO GHO ingestion added 1,005 reference resistance values spanning 103 countries and 2016–2023.

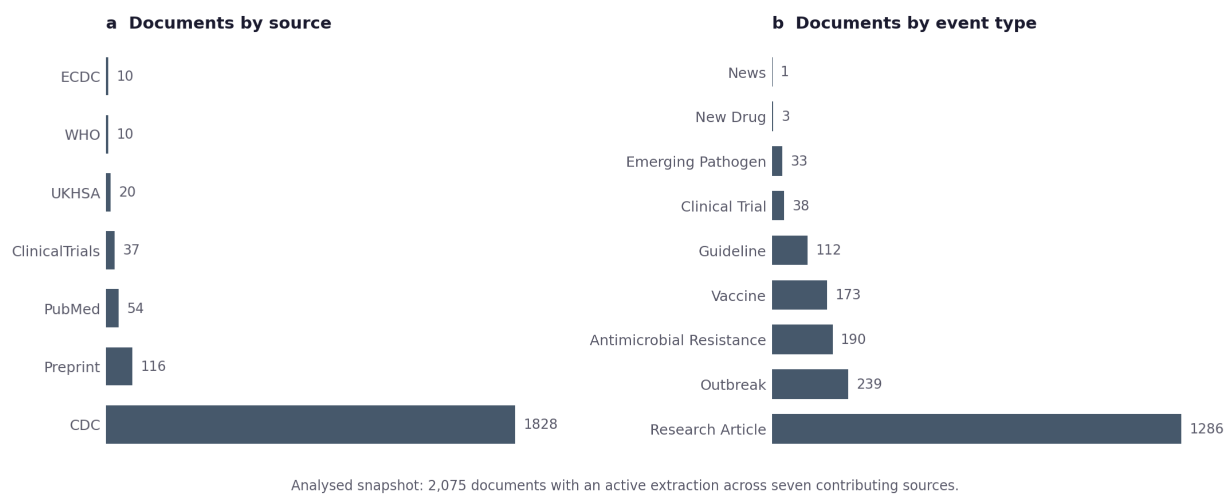

Figure 3. Composition of the analysed corpus. (a) Documents by source; official alerts, literature, preprints, and trials contribute at very different volumes. (b) Documents by extracted event type. Counts are for the 2,075 documents with an active extraction.

3.2 Extraction and knowledge graph

Structured extraction reproduced expected AMR concepts across pathogens, resistance mechanisms, and antibiotics (Figure 4), and, for the LLM extractor, recovered fine-grained resistance alleles absent from the baseline dictionary. Graph queries returned interpretable chains — for example, around carbapenem resistance the platform linked carbapenemase genes (OXA-48, KPC, NDM) to the Enterobacterales and non-fermenters that carry them, each edge traceable to the underlying documents (Figure 5).

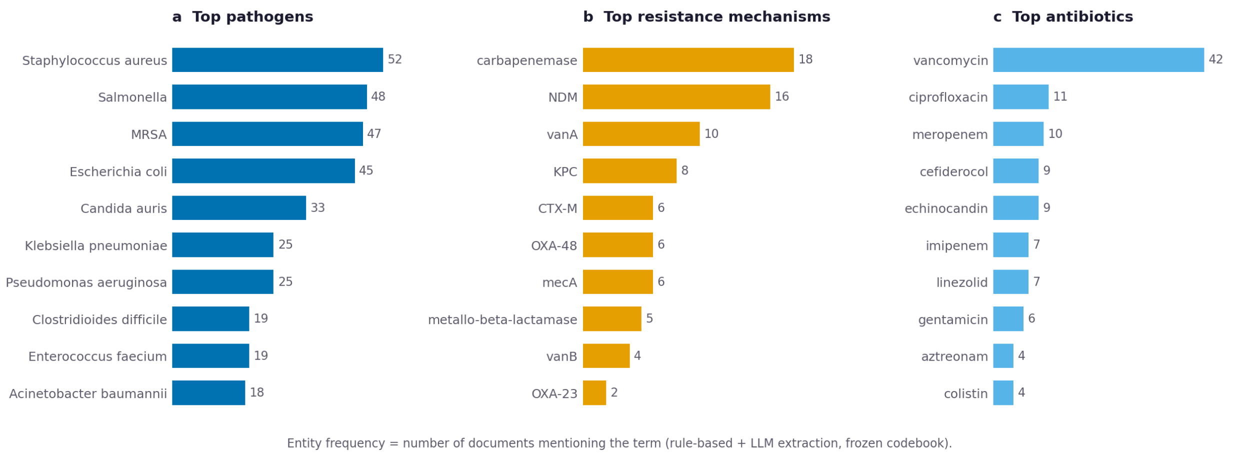

Figure 4. The AMR entity landscape VIGIL extracts. Most-mentioned (a) pathogens, (b) resistance mechanisms, and (c) antibiotics, by number of documents. The priority phenotypes and carbapenemase genes that dominate current AMR concern are surfaced directly from open-source text.

3.3 Preliminary validation

Across the 43 countries with both a platform AMR signal and a WHO GHO MRSA value, raw AMR-document volume was essentially uncorrelated with measured resistance (Spearman $\rho = 0.07$), and a publication-normalised AMR share recovered only a weak positive association ($\rho = 0.16$; Figure 6) — consistent with document volume reflecting research output rather than disease burden, and with normalisation partially recovering signal. The country-level signal is sparse: most countries contribute only a handful of AMR-tagged documents, so these are proof-of-concept results that are expected to sharpen as the corpus accumulates. A complementary outbreak analysis — which deliberately separates official-alert outcomes from literature-derived features — likewise found that literature signal did not track where outbreaks were officially reported, which concentrated in lower-resource, lower-literature settings. Even at this stage, the platform makes the surveillance gap explicit and measurable rather than hidden.

4 Discussion

VIGIL complements, rather than duplicates, existing open-source epidemic-intelligence systems: where those emphasise outbreak-event detection, VIGIL emphasises **structured, AMR-specific extraction, a knowledge graph, and validation against reference resistance data**, all fully open and reproducible. Its most striking early observation — that raw open-source literature signal is uncorrelated with measured resistance and inversely associated with where outbreaks are officially reported — is itself a substantive finding: naïve text-volume indicators would misrank global risk, and any credible early-warning model must correct for reporting and publication bias. By quantifying this gap against WHO reference data, VIGIL turns a known caveat into a measurable target.

The platform is also designed as research infrastructure. Because it stores both rule-based and LLM outputs with full provenance and ships a frozen annotation codebook, it directly enables a clinician-annotated benchmark of open LLMs for AMR extraction, and its accumulating time series supports early-warning modelling with lead-time evaluation against GLASS. Its Asia-Pacific focus addresses a region under-represented in existing tools.

More broadly, we believe **validation-first surveillance should become a fundamental design principle for the next generation of infectious-disease intelligence systems**: as automated extraction and large language models make it trivial to *generate* signals from open text, the scarce and decisive capability is no longer generation but the standing infrastructure to validate, calibrate, and continuously re-check those signals against reference truth (Figure 2). VIGIL is one concrete instance of that principle; we hope it is among the first of many.

5 Limitations

The corpus is young and unevenly dated (Figure 7), so trend, forecasting, and novelty analytics are preliminary and strengthen with accumulated history. Text-derived geolocation reflects where topics are reported or studied, not con-

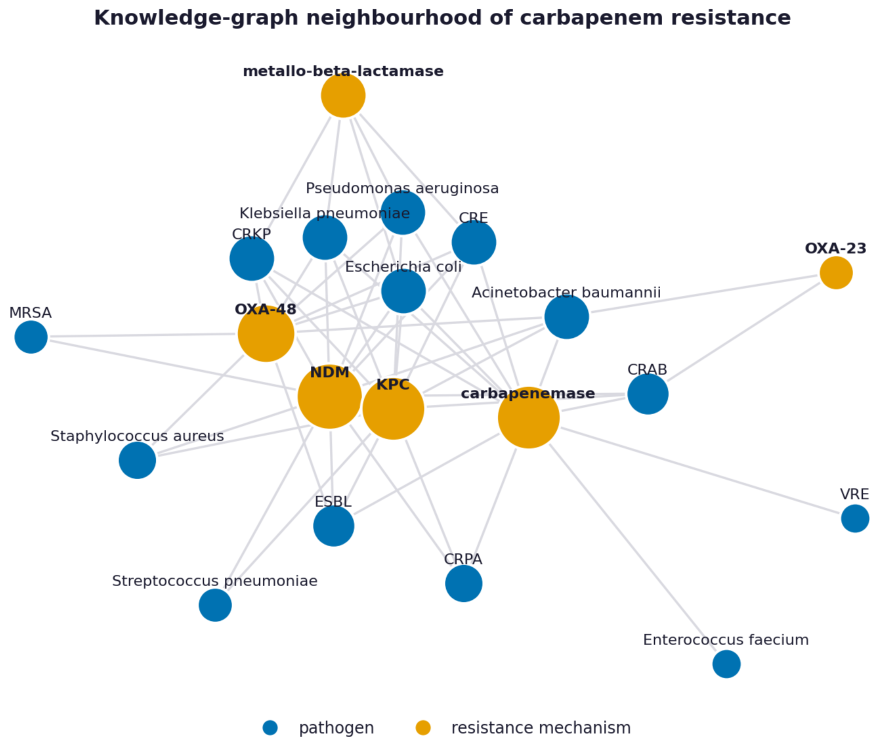

Figure 5. Knowledge-graph neighbourhood of carbapenem resistance in the corpus: carbapenemase genes (orange) and the pathogens (red) they co-occur with; node size scales with degree. Every edge is traceable to source documents.

firmed case counts, and is subject to reporting and publication bias (which the platform surfaces rather than conceals).

Reference coverage is incomplete: WHO GHO does not list some territories, including Taiwan (a listing limitation of the reference source, not a platform merge), and lacks values for several countries on these indicators. LLM extraction is currently constrained by free-tier throughput. Finally, the extraction benchmark awaits full clinician annotation and a second annotator for reliability estimation.

6 Conclusion

VIGIL is an open, validation-first platform that converts open-source infectious-disease and AMR text into structured, citable intelligence, and a foundation for a program of methodological studies. It is freely available, runs and updates at no cost, and is archived with a DOI for reuse and citation.

Declarations

Data and code availability: all code and the annotated schema are available on GitHub and archived on Zenodo (DOI: 10.5281/zenodo.21263882); data derive from public open-source APIs. **Funding:** none. **Competing interests:** none declared. **Ethics:** the platform uses only open-source, aggregate/public data and no individual patient data. **Author contributions:** YHW conceived, designed, implemented, and wrote the work.

References

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[2] World Health Organization. Global Antimicrobial Resistance and Use Surveillance System (GLASS). Geneva: WHO. Available from: https://www.who.int/initiatives/glass

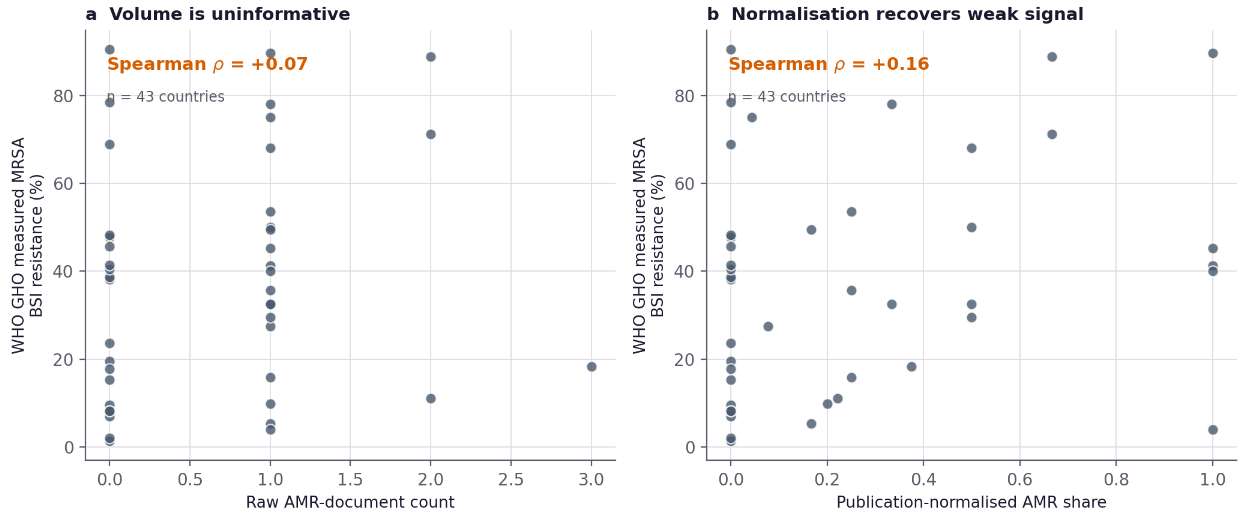

Figure 6. Construct validity against independent reference data, across 43 countries. (a) Raw AMR-document count versus WHO GHO measured MRSA bloodstream-infection resistance is essentially uncorrelated (Spearman $\rho = 0.07$); (b) a publication-normalised AMR share recovers only a weak positive association ($\rho = 0.16$). Text volume alone does not track measured resistance — a quantification of the global surveillance gap.

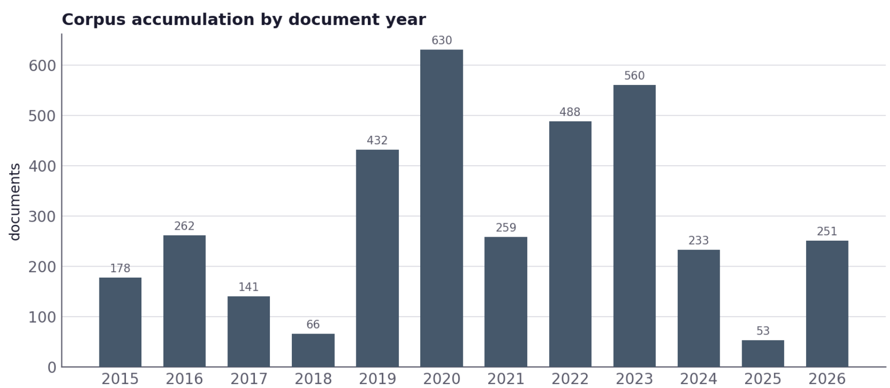

Figure 7. Documents by document year in the corpus. Coverage is concentrated in recent years and accumulates daily; longitudinal analytics gain power as the historical series lengthens.

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Appendix A. Live instance

The infrastructure runs as a public, daily-updated web instance (streamlit.app), which exposes every module in Table 2 together with the search, knowledge-graph, heatmap, annotation, and benchmark tools (Figure 8).

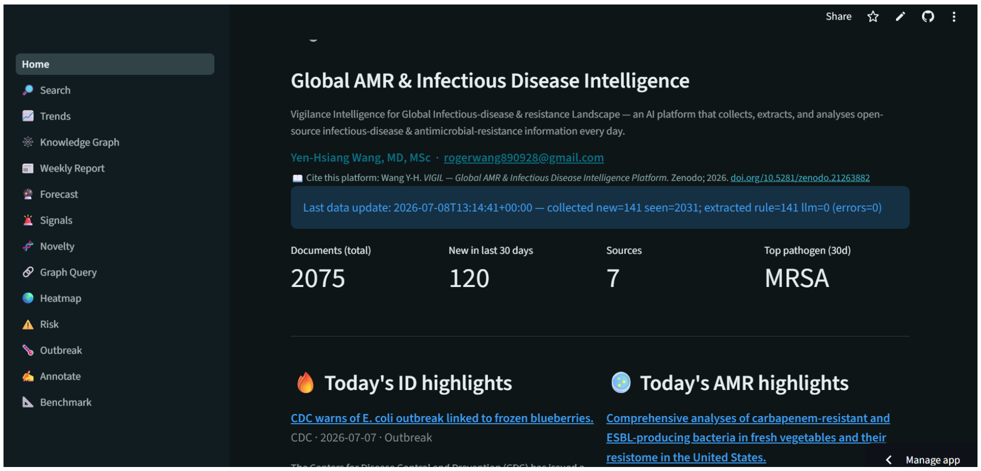

Figure 8. The VIGIL public live instance. Shown for reproducibility and reuse; it is the running realisation of the infrastructure in Figure 1, not a contribution in itself.